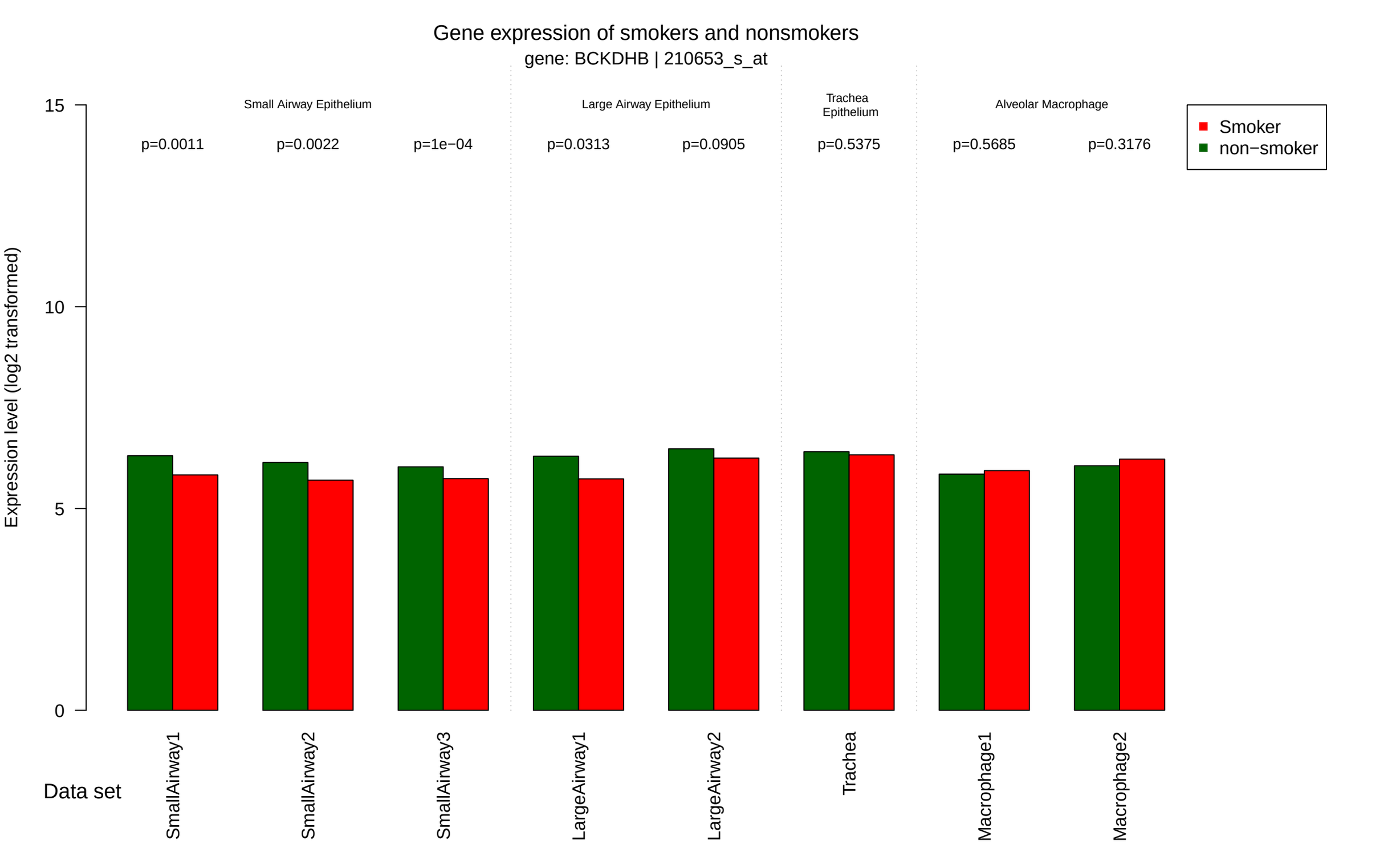
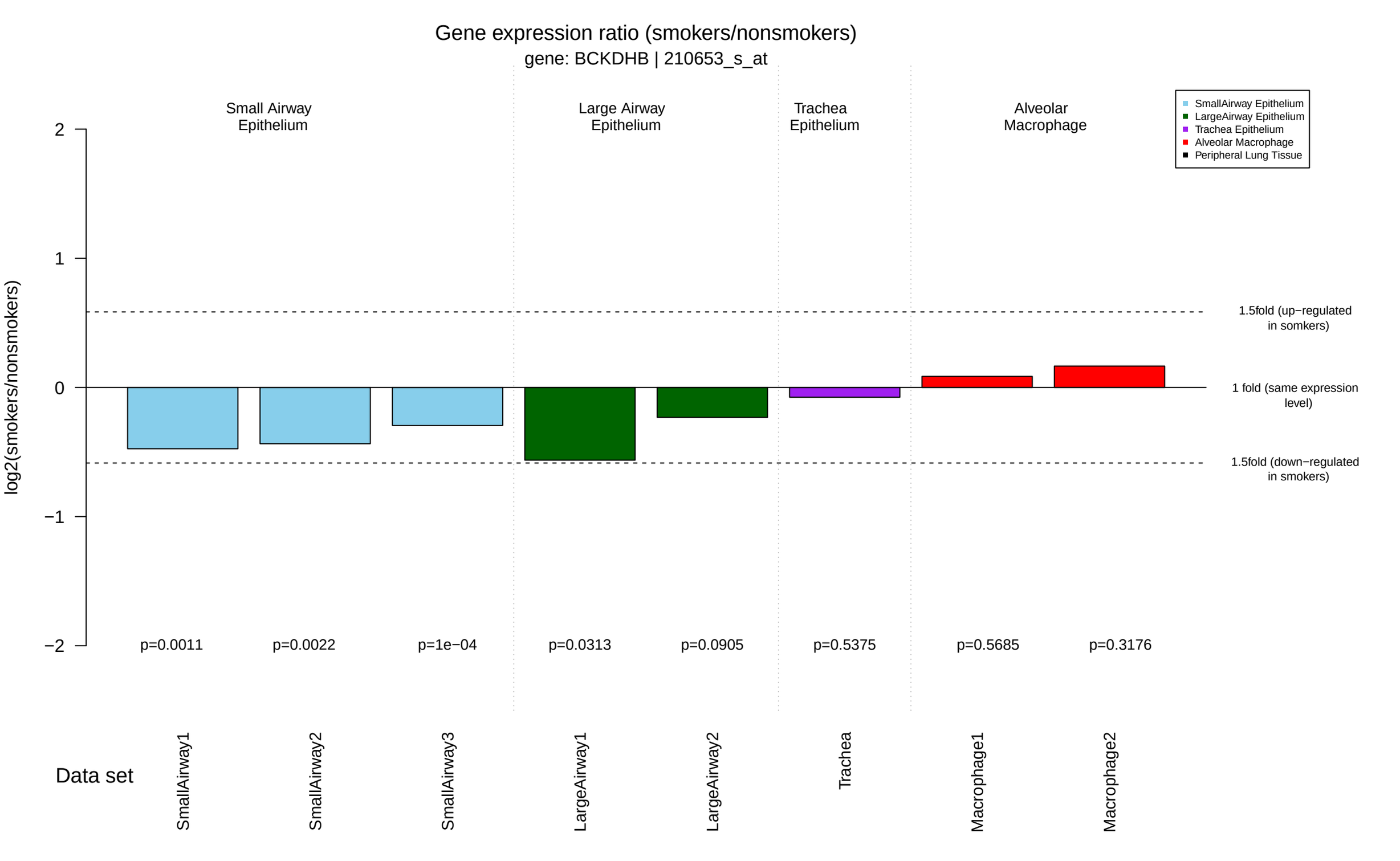
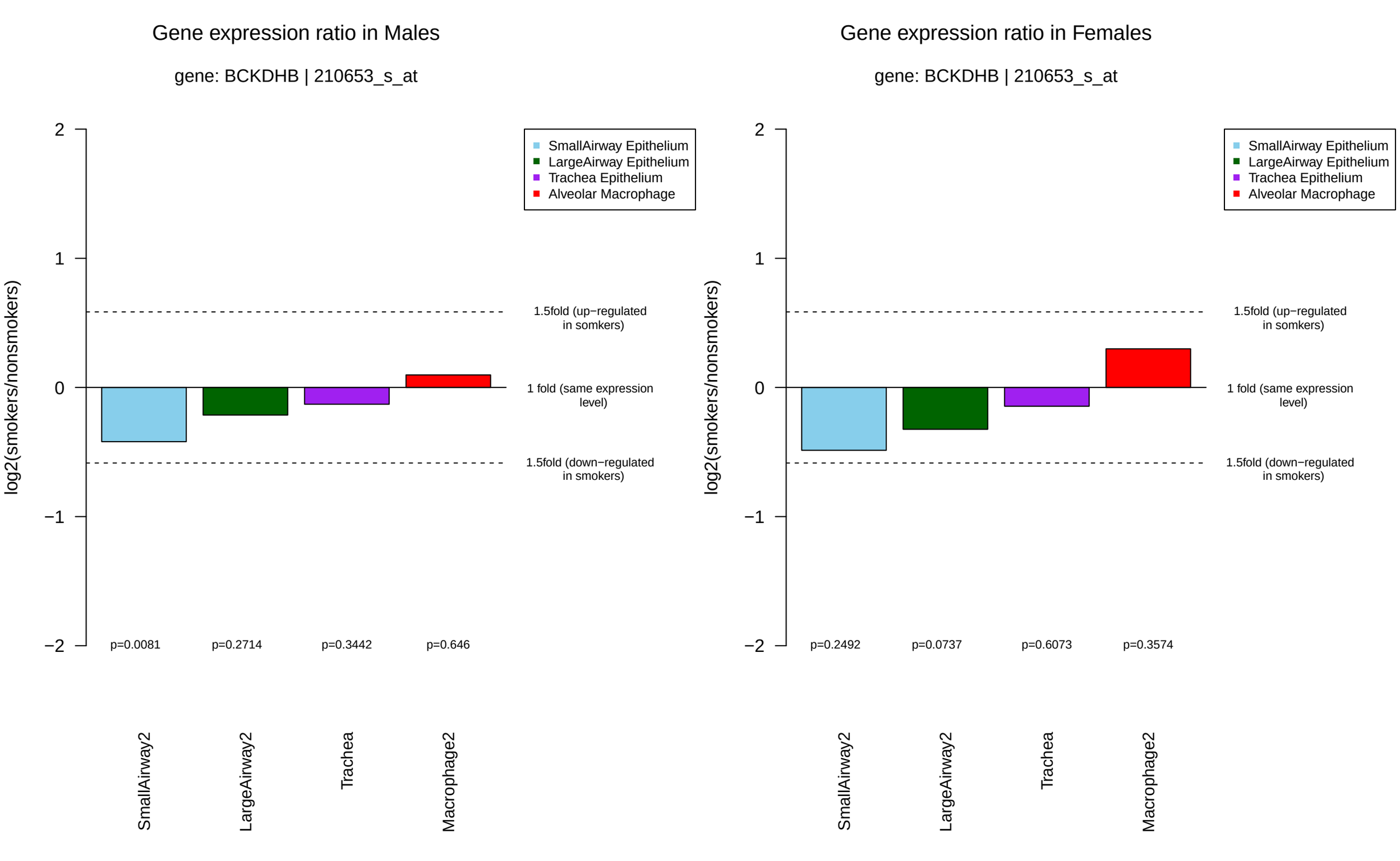
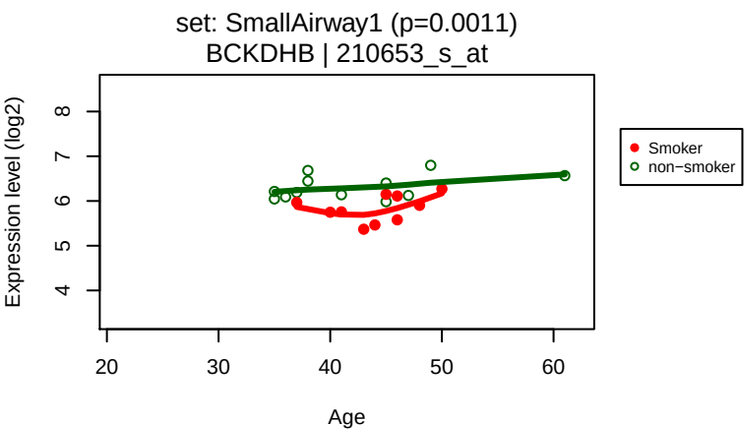
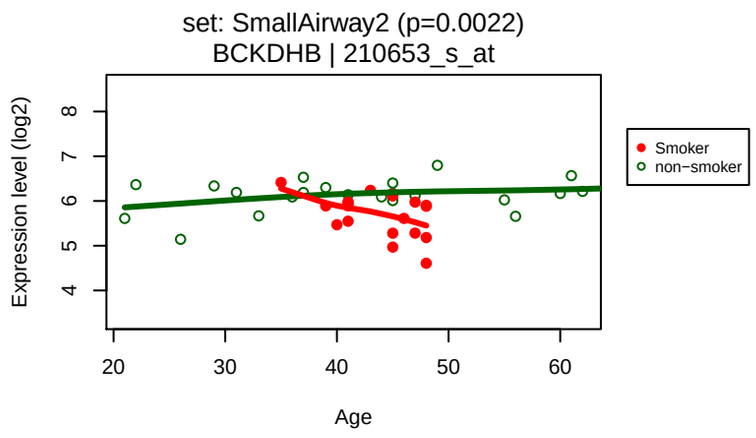
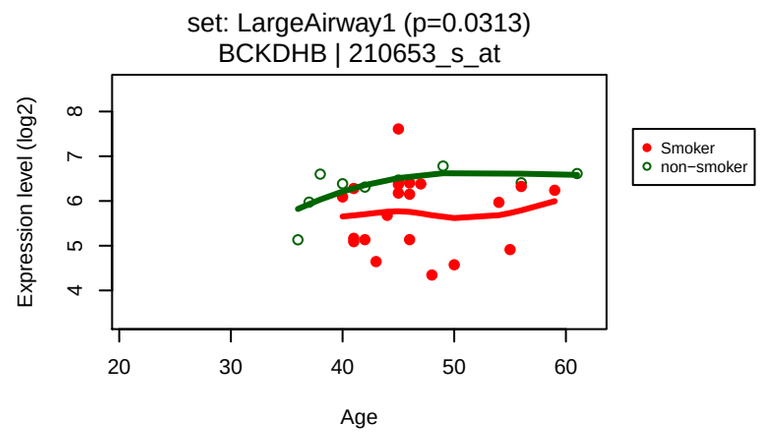

Gene expression of smokers and nonsmokers
gene: BCKDHB | 210653_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0011
p=0.0022
p=1e−04
p=0.0313
p=0.0905
p=0.5375
p=0.5685
p=0.3176
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

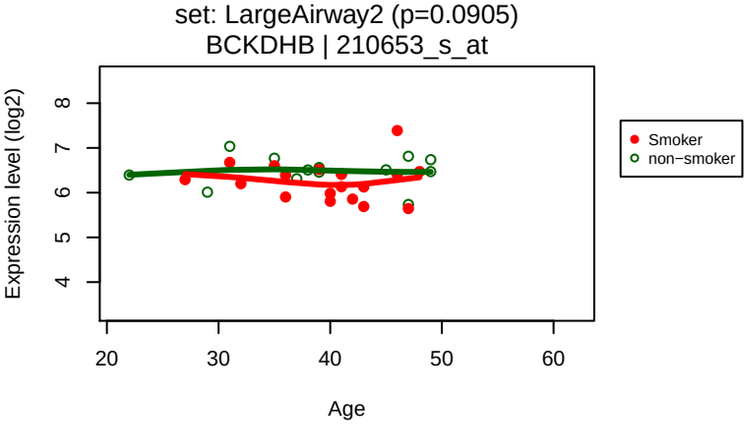

Gene expression ratio (smokers/nonsmokers)
gene: BCKDHB | 210653_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0011
p=0.0022
p=1e−04
p=0.0313
p=0.0905
p=0.5375
p=0.5685
p=0.3176
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

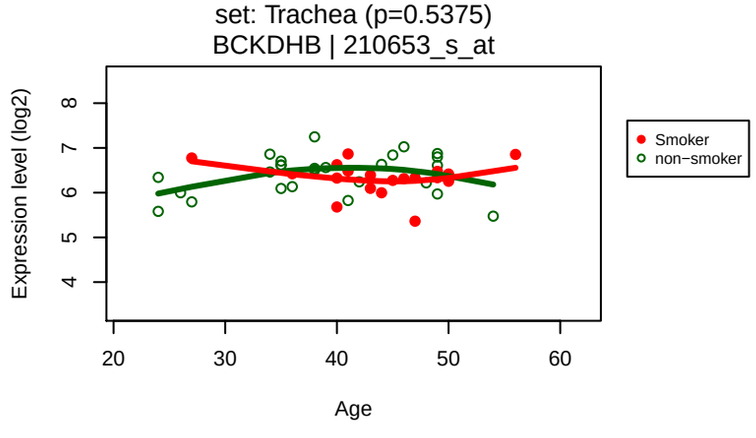

Gene expression ratio in Males
gene: BCKDHB | 210653_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0081
p=0.2714
p=0.3442
p=0.646
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCKDHB | 210653_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2492
p=0.0737
p=0.6073
p=0.3574
SmallAirway2
LargeAirway2
Trachea
Macrophage2

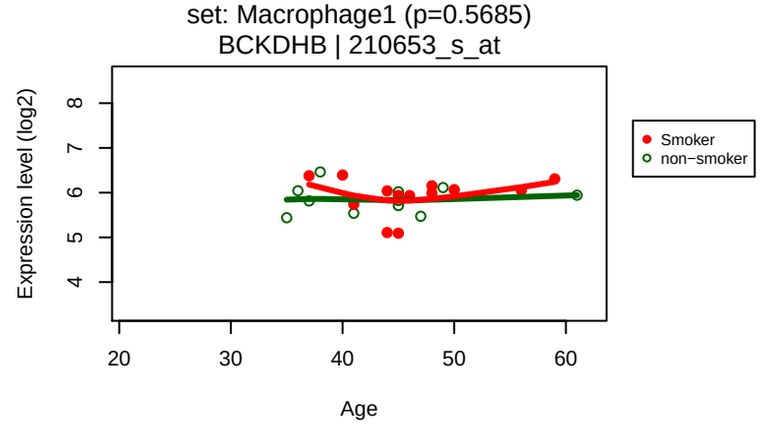

set: SmallAirway1 (p=0.0011)
BCKDHB | 210653_s_at
Expression level (log2)
Age
Smoker
non−smoker

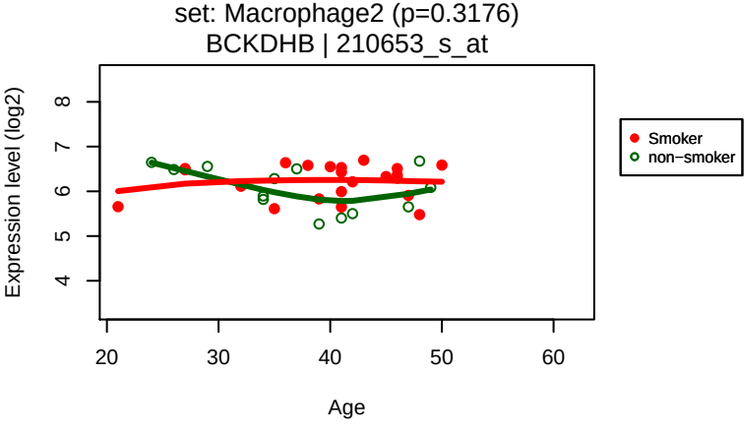

set: SmallAirway2 (p=0.0022)
BCKDHB | 210653_s_at
Expression level (log2)
Age
Smoker
non−smoker

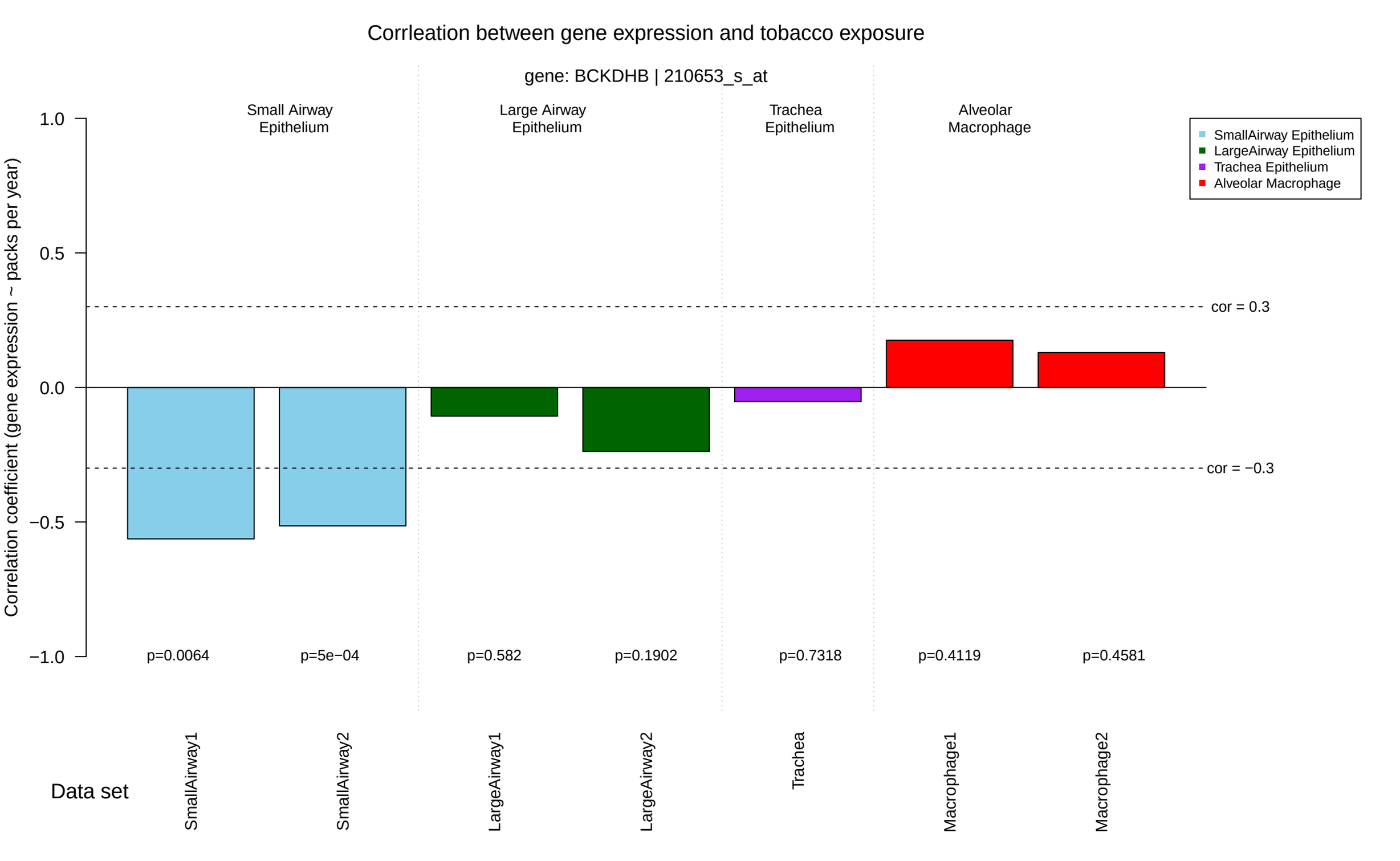

set: LargeAirway1 (p=0.0313)
BCKDHB | 210653_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0905)
BCKDHB | 210653_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5375)
BCKDHB | 210653_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5685)
BCKDHB | 210653_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3176)
BCKDHB | 210653_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCKDHB | 210653_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0064
p=5e−04
p=0.582
p=0.1902
p=0.7318
p=0.4119
p=0.4581
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2